EMS690U Integrated Design Project

Assessment 4: Individual Detailed Design Report

Anatomical Core: Design, Manufacture, and Integration of the Fracture Immobilisation Trainer

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22 Apr. 26

Abstract

This report presents the design, manufacture and testing of the anatomical core for the Interactive Fracture Immobilisation Trainer, a high-fidelity cast-application simulator targeting distal radius (Colles') fractures. The core is the physical integration platform for all sensing, haptic and fluidic subsystems. It covers a full upper limb from humerus to fingertips with a 3D-printed skeletal structure with internal routing channels, TPU ligament strips at articulating joints, and a dual-durometer silicone soft-tissue system (Smooth-On Ecoflex 00-30 skin over Dragon Skin 30A muscle). Multi-part moulds were designed in Blender, printed in PLA and filler-primed. A cantilever deflection study of four filaments identified PETG-CF as the optimal skeletal material on elastic-stability grounds and this is the production specification; the prototype uses PLA and ASA-CF owing to filament constraints. Preliminary tensile testing confirmed that the silicones behaved consistently with datasheet specifications. The fracture site is a clean transverse break in the distal radius, tracked via hall-effect sensors for real-time alignment feedback. At an estimated material cost below £500, this work demonstrates that anatomically accurate, multi-layered training models can be produced using accessible digital fabrication methods.

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List of Symbols

Symbol	Description	Unit
σ	Stress	MPa
ε	Strain	mm/mm
E	Young's Modulus	MPa
F	Applied Force	N
A	Cross-sectional Area	mm^2
t	Thickness	mm
Shore	Shore Hardness	Non-SI Scale (0-100), (00, A, D)

1. Introduction

Distal radius fractures account for approximately 17.5% of all fractures presenting to UK emergency departments (Jennison and Brinsden, 2019), making correct cast application a foundational clinical skill. Training opportunities on real patients are increasingly constrained by ethical concerns and patient apprehension about trainee involvement (Okuda et al., 2009), yet the consequences of improper casting are serious: thermal burns from exothermic plaster reactions, pressure sores from inadequate padding, and compartment syndrome from excessive tightness can all lead to significant morbidity (McGraw-Heinrich, Wall and Rosenfeld, 2025).

This project addresses these issues through the Interactive Fracture Immobilisation Trainer, a physical forearm simulator integrating sensing, haptic feedback, a pulsatile flow system, and a guided user interface into a single platform. Existing commercial trainers such as the Limbs & Things Colles' Fracture Trainer (~£2,000)(Limbs & Things, 2019) are passive: they provide a realistic substrate for practice but deliver no objective assessment, so feedback depends on an observing instructor. The present device closes this loop via embedded sensors that quantify fragment alignment, cast pressure distribution, and setting temperature for immediate review, at a small fraction of the commercial cost. This individual report covers the anatomical core workstream: the skeletal structure, joints, soft-tissue layers, mould design, material selection, and fracture-site engineering that together form the chassis on which all other subsystems integrate.

The central aim is a functional anatomical artefact simultaneously anatomically faithful enough to preserve the palpation cues trainees rely on, mechanically robust enough to survive repeated cast-application cycles, and modular enough to host the sensing and actuation subsystems developed by other team members. These three requirements are in tension: higher fidelity means softer, more fragile materials, while durability and modularity push towards stiffer, simpler parts. The design decisions in this report reflect the trade-offs; the cantilever study in Section 5.3 provides the quantitative basis for the skeletal material selection.

The remainder of this report is organised as follows. Chapter 2 provides a literature review of anatomical modelling and silicone tissue simulation. Chapter 3 details the design methodology and tools employed. Chapter 4 presents the detailed design of each component. Chapter 5 covers manufacture and testing. Chapter 6 provides a critical discussion including ethical and environmental considerations. Finally, Chapter 7 draws conclusions and identifies future work.

2. Literature Review

2.1 Simulation in Medical Education

The shift towards simulation-based medical education is well established. Okuda et al. (2009) showed that high-fidelity simulators improve both knowledge retention and procedural confidence in medical trainees. Cook et al. (2011) subsequently carried out a systematic review and meta-analysis of over six hundred studies, linking technology-enhanced simulation to large effects on knowledge and skills and moderate effects on patient outcomes. Together these studies establish simulation as a mainstream intervention, and emphasise that passive static models offer limited value compared with systems providing active feedback.

The specific case of cast application has received less attention than surgical procedures despite the well-documented clinical consequences of poor technique. Moktar et al. (2014) developed a synthetic short-arm cast-application simulator and validated it against an OSATS checklist generated by Delphi consensus among orthopaedic surgeons and cast technologists, demonstrating construct validity by differentiating medical students, residents and fellows on both procedural and global-rating scores. Their platform is, however, passive: assessment depends on expert observation rather than real-time feedback from the model. This identifies the gap the present project addresses: a cast-application trainer delivering immediate objective feedback on technique rather than relying on post-hoc expert review.

2.2 Physical Fidelity in Fracture Models

The importance of multi-layer tissue construction for realistic palpation has been demonstrated by several groups. A study by Dixon et al. (2021) developed a 3D-printed simulator for closed reduction of distal radius fractures using 3D printed bones. While the mechanical reduction was effective, the authors acknowledged that tactile realism was limited by the uniform material properties. Favier et al. (2021) argue that haptic fidelity is the key missing dimension in most current surgical simulators, and that accurate reproduction of tissue-specific mechanical responses, rather than visual appearance alone, is what determines whether practised skills transfer safely to patients. This directly motivates the dual-durometer approach adopted in this project.

Quantitatively, the rationale is supported by direct measurements of human forearm tissue. Iivarinen et al. (2011) used manual indentation coupled with B-mode ultrasound and a layered hyperelastic finite-element model to separate the mechanical contributions of skin, adipose tissue and muscle, recovering elastic moduli of ~210 kPa for skin and 1.9 kPa for adipose tissue at rest, a difference of two orders of magnitude. They further showed the indentation response was 9 to 16 times more sensitive to skin mechanics than to adipose or muscle, meaning realistic skin mechanics dominate the tactile impression during palpation. This justifies a stiff outer skin layer (Ecoflex over Dragon Skin) rather than a single bulk silicone, and sets a target compliance ordering that the dual-durometer system approximates.

2.3 Material Selection for Tissue Simulation

Platinum-cure silicone rubbers are the industry standard for soft tissue simulation due to their biocompatibility, tear resistance, and tuneable Shore hardness. Smooth-On’s Ecoflex series (Shore 00-30 to 00-50) is widely used for skin analogues due to its high elongation at break (900%) and low viscosity for detailed mould reproduction (Smooth-On, 2023b). For deeper tissue requiring greater elastic recoil, Dragon Skin 30 (Shore A 30) provides a stiffer matrix that better replicates the resistance encountered during palpation of bony landmarks through muscle (Smooth-On, 2023a). The combination of these two materials at different durometers allows the non-linear stress-strain response of human soft tissue to be approximated, where skin exhibits high compliance at low strains but stiffens under load, and underlying muscle provides firm elastic support.

2.4 3D Printing for Skeletal Analogues

Fused deposition modelling (FDM) is the most accessible method for producing anatomical bone models. PLA is commonly used for ease of printing and fine detail but its brittleness and tendency to delaminate along layer lines make it unsuitable for repeated mechanical handling (Dizon et al., 2018). PETG offers a better balance of ductility, layer adhesion and chemical resistance, with elongation at break ~18–23% vs PLA's 3–6% (Dizon et al., 2018). ABS is tougher but was rejected on emission grounds: Wojtyła, Klama and Baran (2017) identified styrene and other VOCs released during ABS printing at levels warranting an enclosed chamber or controlled ventilation, neither compatible with the available workshop. TPU at 95A Shore hardness provides the flexibility needed for ligament analogues while remaining printable on standard FDM hardware.

2.5 Pain Physiology and Sensing Requirements

Mitchell, Majuta and Mantyh (2018) describe the periosteum as the primary source of acute fracture pain due to its dense nociceptor innervation; mechanical distortion of the fragments activates these fibres with pain proportional to displacement. Mach et al. (2002) mapped the sensory and sympathetic innervation of the mouse femur by immunohistochemistry and confocal microscopy, showing that the periosteum is the most densely innervated of the three bone compartments per unit volume, with fibres organised as a branched meshwork adapted to mechanical distortion rather than associated with blood vessels. This explains why small relative displacements at a fracture site produce disproportionately large pain responses, and why mechanical stabilisation (the goal of casting) is so strongly analgesic.

This understanding informed the decision to instrument the fracture site with displacement sensors (Hall-effect magnetometers) rather than relying solely on surface force measurements. The quantity the trainee is being taught to control is mechanical displacement at the periosteum, so the sensing modality most closely aligned with the pain-generating physiology is direct measurement of fragment displacement. The sensing workstream developed the sensor integration strategy, while this workstream was responsible for providing appropriate mounting geometry and internal routing within the skeletal structure.

3. Design Methodology

3.1 Design Tools

The anatomical core used a combination of organic and parametric CAD tools. Blender 4.5 handled organic geometry (bone sculpting, soft-tissue outer forms, mould cavities), chosen for its sculpting and Boolean capabilities on complex anatomical shapes that are hard to model parametrically. Onshape was used for joint components (TPU ligament strips) and fastener geometry where parametric dimensional control was required. Slicing was performed in Orca for PLA, PETG (CF), ASA (CF) and TPU.

3.2 Design Philosophy

The design followed an iterative, modular philosophy. The skeleton was designed first as the structural core, with internal channels and mounting features defined early to establish interfaces with the other workstreams. Soft-tissue moulds were designed around the finalised bone geometry, with silicone casting last. Regular consultation with the flow, sensing and haptic-feedback teammates ensured interface dimensions and tolerances were agreed before manufacture.

3.3 Anatomical Reference and Scaling

The skeletal geometry was sculpted in Blender from photographic reference of real human bones, scaled to average adult forearm dimensions. The forearm length (radiale to stylion) was set at 255 mm from 50th-percentile adult anthropometric data (NASA, 1978). The model spans humerus to fingertips, providing a complete upper limb for realistic cast-application training that extends beyond the fracture site.

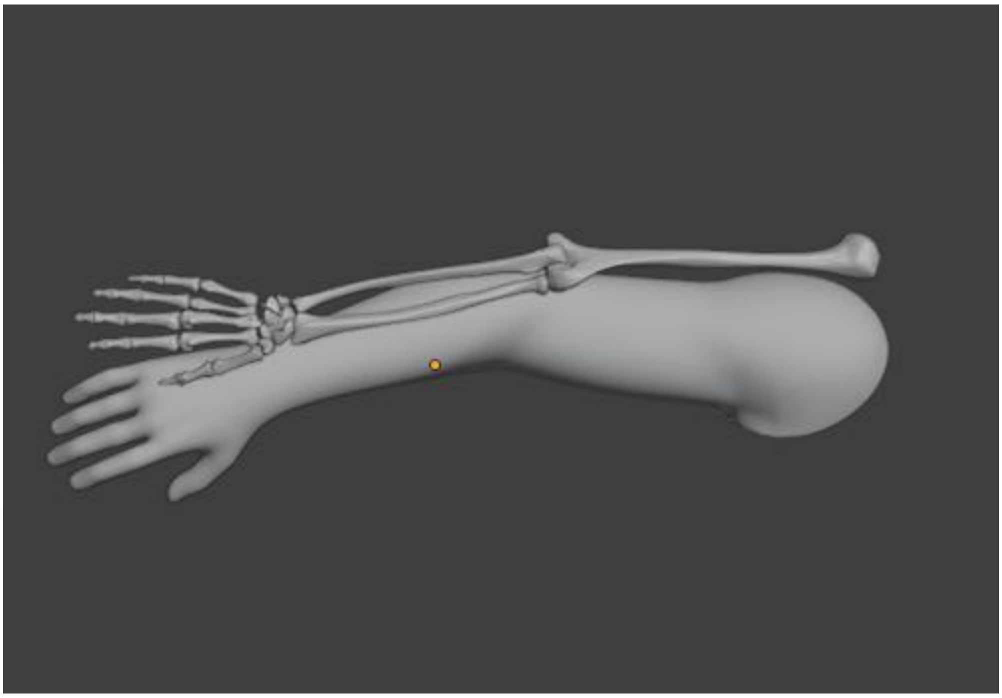

Figure 1 – Arm and skeleton full model designed in Blender

3.4 Concept Selection

Material and architectural choices for the anatomical core were made using a weighted decision matrix approach. Evaluation criteria were drawn directly from the functional requirements in the design brief and the literature reviewed in Chapter 2. Two decisions are documented in full in this subsection: the selection of a skeletal filament, and the pairing of silicones for the dual-durometer soft tissue system. Other decisions (mould split lines, channel geometry, mounting features) were made iteratively during detailed design and are justified inline in Chapter 4.

For the skeletal filament, four materials were scored from 1 (worst) to 5 (best) on five weighted criteria. Weights were set a priori from the priorities in Chapter 2: elastic stability was weighted most heavily because cast application is a cyclic long-duration load, so dimensional stability across many cycles matters more than peak strength. Workshop compatibility eliminates ABS directly because the available shared workspace cannot host an enclosed print chamber (Wojtyła, Klama and Baran, 2017). The cantilever study in Section 5.3 was commissioned to generate the evidence for the elastic-stability column.

Table 1 – Weighted decision matrix for skeletal filament selection

Criterion	Weight	PLA	PETG-CF	ASA-CF	ABS
Elastic stability across load range	0.35	2	5	3	2
Working-stress margin above trainer load	0.20	3	4	4	3
Printability on Qidi Q2 (temperature, adhesion)	0.15	5	4	3	3
Filament and print cost per full-limb set	0.15	5	3	2	4
Workshop compatibility (no enclosure required)	0.15	5	5	4	1
Weighted total (out of 5.0)	**1.00**	**3.55**	**4.35**	**3.20**	**2.50**

PETG-CF wins at 4.35, driven by its dominant elastic-stability score. PLA (3.55) is held back on the most heavily-weighted criterion; ASA-CF (3.20) could not be fully confirmed on elastic stability because its first cantilever point was at the ruler resolution limit; ABS (2.50) is eliminated on workshop compatibility. PETG-CF is therefore the production specification; the prototype PLA/ASA-CF choices in Section 5.1 are a filament-constrained fallback, not a change in ranking.

A similar matrix was built for the silicone architecture, comparing the adopted dual-durometer (Ecoflex over Dragon Skin) system, a single bulk Ecoflex option, and a polyurethane foam inner with silicone skin. Multi-layer compliance reproduction was weighted most heavily because it is the key requirement established by Iivarinen et al. (2011) and Favier et al. (2021). Mould release compatibility was included because polyurethane foam inhibits the cure of platinum-cure silicones without a barrier layer.

Table 2 – Weighted decision matrix for silicone soft-tissue architecture

Criterion	Weight	Dual-durometer (Ecoflex + Dragon Skin)	Single silicone (Ecoflex only)	PU foam + silicone skin
Multi-layer compliance reproduction	0.40	5	1	4
Ease of casting / workshop feasibility	0.20	4	5	2
Material cost per artefact	0.15	3	5	4
Mould release compatibility (platinum cure)	0.15	5	5	2
Biocompatibility for repeated skin contact	0.10	5	5	4
Weighted total (out of 5.0)	**1.00**	**4.50**	**3.40**	**3.30**

The dual-durometer system wins at 4.50 as the only option scoring well on the highest-weighted criterion. Single-silicone (3.40) cannot reproduce the two-order-of-magnitude compliance difference between skin and adipose tissue identified by Iivarinen et al. (2011), failing the central requirement. The PU-plus-silicone option (3.30) can in principle layer compliance but the cure-inhibition process penalties and reduced biocompatibility at damaged skin edges outweigh the small cost saving. The dual-durometer architecture is adopted.

4. Detailed Design

4.1 Skeletal Structure

The skeleton comprises individually printed bones for the humerus, radius, ulna, carpals, metacarpals and phalanges, each sculpted in Blender from photographic reference with anatomical landmarks (radial styloid, ulnar head, olecranon) preserved for palpation through the silicone layers. The radius is designed as two separate pieces to create the Colles' fracture at the distal metaphysis, with a clean transverse break geometry allowing dorsal displacement of the distal fragment to replicate the characteristic "dinner fork" deformity.

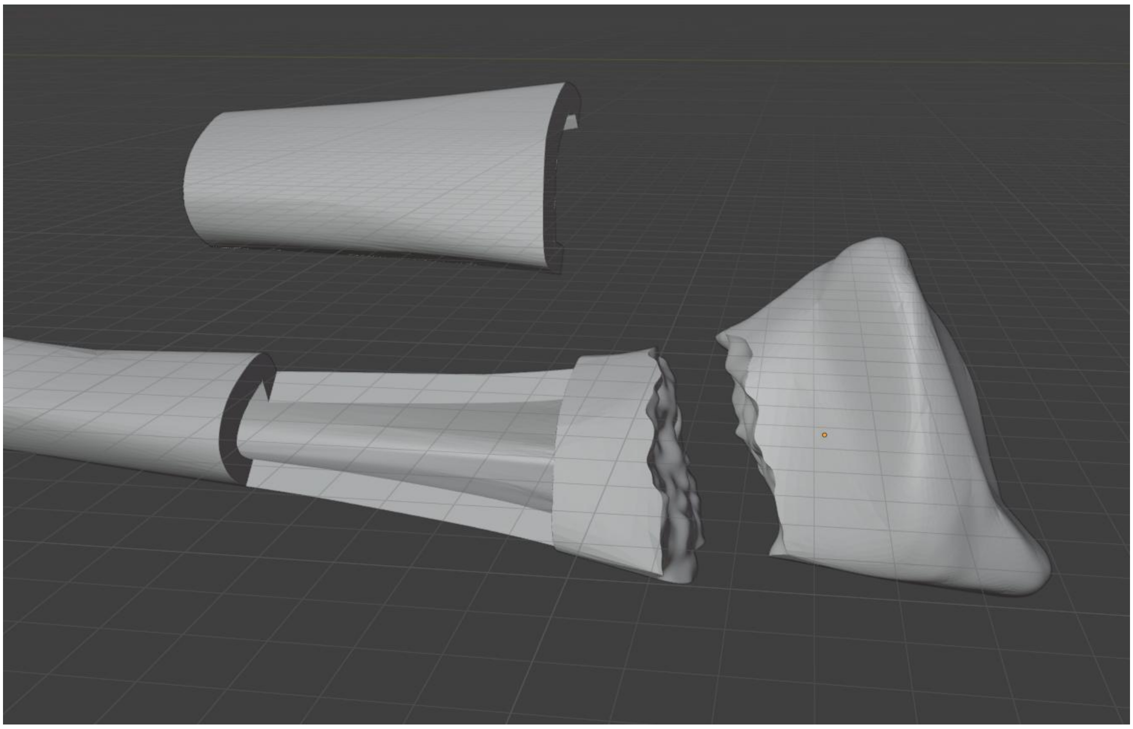

Figure 2 – Radius model in Blender showing bone geometry, fracture, and hollow routing channel

Each bone carries a hollow central channel of approximately 10 mm diameter for sensor wiring and flow-system tubing. This diameter was chosen to allow maximum space for two 22-AWG sensor wires as well as other wires to connect to the interface while preserving adequate cortical wall thickness in narrower sections (notably the distal radius at the fracture site, where outer diameter is ~18 mm). Heat-set M3 brass inserts in the bone ends receive the TPU ligament screws to join to adjacent bones.

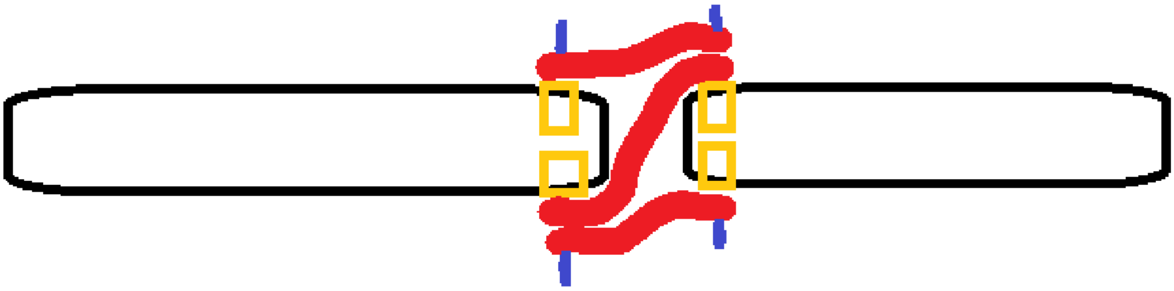

Figure 3 – Schematic of a finger showing two bones with TPU strips and screw fixation points

4.2 Joint Design

Articulating joints were implemented at the radiocarpal (wrist), radioulnar and elbow positions. Early prototyping explored a "cup and knuckle" geometry in a TPU socket matrix but this was abandoned due to manufacturing complexity and inconsistent range of motion. The final design uses simple TPU 95A strips secured to adjacent bones via M3 screws into heat-set brass inserts. This proved more reliable and easier to tune: strip width, thickness and attachment spacing control range of motion, while TPU 95A flexibility permits the full physiological ranges of wrist flexion/extension (~80°/70°) and forearm pronation/supination.

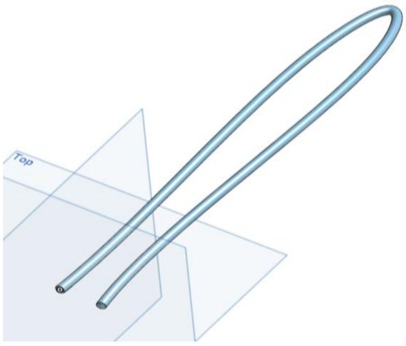

Figure 4 – CAD model of the U-shaped flow tubing designed to route between radius and ulna vertically to maximise space usage while providing adequate feel for trainees

4.3 Soft Tissue System

4.3.1 Muscle Layer (Dragon Skin 30A)

The muscle layer is cast in Smooth-On Dragon Skin 30A, selected for the firm elastic recoil characteristic of palpating through muscle to identify bony landmarks. It is cast in sections corresponding to major anatomical compartments (forearm flexors, extensors, and thenar/hypothenar eminences), each in its own mould piece.

Internal to the muscle layer, several functional features were incorporated. A channel of approximately 4.76 mm internal diameter runs longitudinally between the radius and ulna in a U-shaped loop, oriented vertically so that one branch sits close to the dorsal surface where a palpable pulse can be simulated. This channel accommodates the silicone tubing from the flow pumping system. A shallow recess was designed to seat the TDK PHUA8060 piezoelectric actuator (0.35 mm profile) from the haptic feedback system, positioned near the fracture site. Additional thin channels (approximately 2 mm) provide wire routing from embedded sensors to a central exit point at the proximal end of the forearm.

4.3.2 Skin Layer (Ecoflex 00-30)

The outer skin is Smooth-On Ecoflex 00-30 cast at 2–2.5 mm thickness to replicate the compliance of human skin. It is cast as a single continuous piece in a two-part mould and fitted over the assembled skeleton and muscle like a glove, avoiding seam lines over the casting area.

A silicone release agent spray is applied between the Dragon Skin muscle and the Ecoflex skin to prevent permanent bonding between the two layers. This allows the skin to be peeled off for maintenance, sensor replacement, or inspection of internal components without destroying either layer.

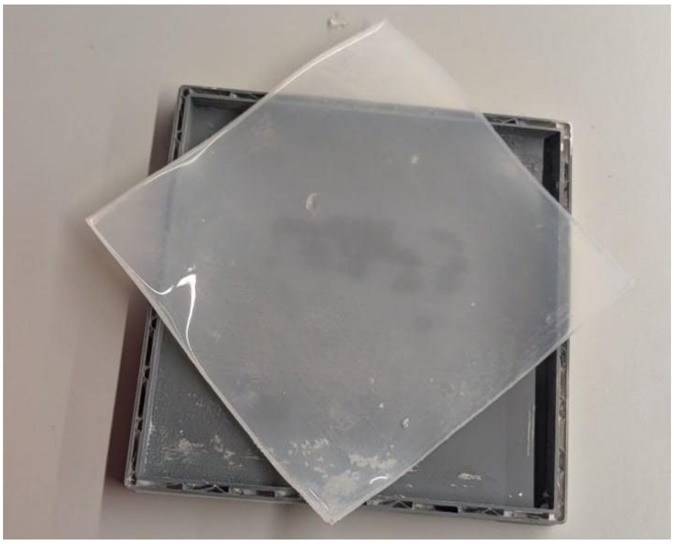

Figure 5 – Test casting of Ecoflex 00-30 in a 150×150 mm square mould, approximately 2.5 mm thickness

4.3.3 Cross-Sectional Architecture

The complete layered structure from inside out is: (1) PETG skeleton with internal wire channels; (2) TPU joint strips at articulation points; (3) Dragon Skin 30A muscle with embedded tubing, sensor pockets and actuator recesses; and (4) Ecoflex 00-30 skin envelope. This layered approach approximates the non-linear stress-strain behaviour of real forearm tissue, where low-force palpation encounters soft skin, moderate force meets muscular resistance, and firm pressure reveals bony landmarks.

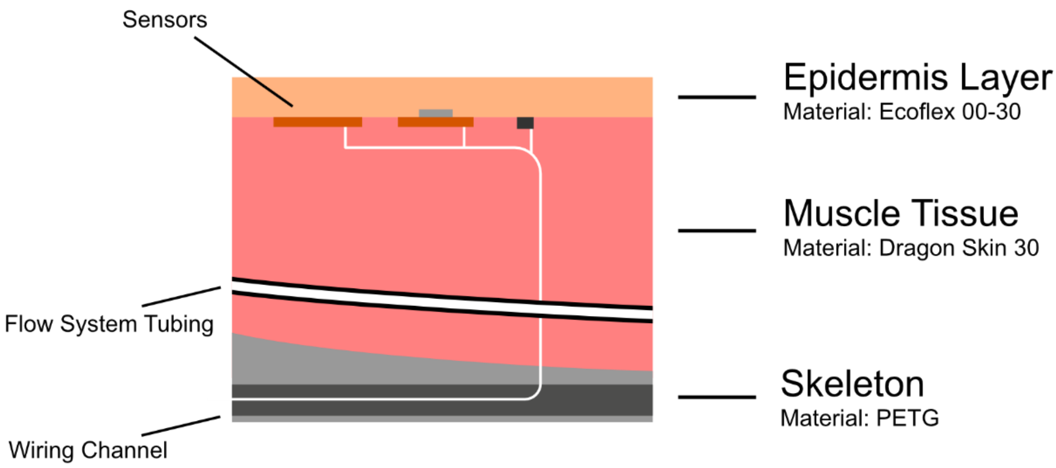

Figure 6 – Model layer diagram created by B. Skulimovskyi

4.4 Mould Design and Manufacture

All silicone moulds were designed in Blender using Boolean difference between the outer arm geometry and the bone/joint assembly. The outer arm form was modelled on Blender using the sculpting feature based off many reference images. Moulds were split into sections not exceeding 250×250 mm to fit the Qidi Q2 print bed, with alignment pins and bolt holes for repeatable multi-part assembly.

Figure 7 – 3D printed moulds: forearm section (orange PLA) and hand section (blue PLA) with support structures printed for the muscular layer

Figure 8 – Hand moulds (grey, post-primed) and forearm section moulds (white PLA) ready for silicone casting for the epidermal layer

The mould surfaces were finished in three stages: sanding with 120-grit paper to remove support marks and layer lines; two coats of filler primer spray; and, for moulds producing the external skin surface, a final 400-grit pass for a smooth transfer to the silicone. Muscle moulds received only the primer, as surface finish on internal layers is non-critical.

Figure 9 – Complete mould set with finished surfaces alongside the core, allowing an epidermal layer to cast when assembled

4.5 Flow System Integration

The flow pumping system requires a closed-loop tubing circuit through the forearm to simulate arterial pulsation. The core accommodates this via a U-shaped channel cast into the Dragon Skin muscle layer: tubing enters at the proximal end, runs distally between the radius and ulna, loops at the wrist and returns. The vertical orientation places one branch immediately beneath the dorsal skin surface so pulsatile flow can be palpated as a simulated radial pulse. The tubing is standard 4.76 mm (3/16”) ID silicone, the smallest readily available bore compatible with the teammate’s centrifugal pump and matching commodity laboratory peristaltic tubing so the assembly can be serviced from stock components.

Figure 10 – 3D-printed PLA prototype of the U-shaped tubing geometry

5. Manufacture and Testing

5.1 3D Printing Parameters

The production specification for all skeletal components is PETG-CF (carbon-fibre reinforced PETG) printed on a Qidi Q2 at 245°C nozzle, 80°C bed, 0.2 mm layer height, 4 perimeter walls, 20% gyroid infill, 50 mm/s. Gyroid was chosen for its near-isotropic response to the multi-axial cast-application loads; 20% density is the minimum that preserved surface print quality in preliminary samples while minimising filament cost. PETG-CF was selected on the evidence of the cantilever study in Section 5.3. The prototype deviates owing to filament and time constraints: most bones were printed in PLA at 230°C and the distal radius in ASA-CF at 270°C, both at 15% infill to conserve filament for the full-limb print. CAD geometry, wall count and infill pattern are unchanged between prototype and production; only filament, infill density, and nozzle temperature differ. Mould components were printed in PLA at 210°C as moulds experience negligible mechanical loading.

5.2 Silicone Tensile Testing

Tensile test samples were cast in 150×150 mm square moulds at controlled thicknesses (Ecoflex 00-30 at 2.5 mm, Dragon Skin 30A at 3.5 mm) and cut into 25×100 mm strips for uniaxial testing.

Preliminary results are presented in Table 3. The manual pull-and-measure rig is not sufficient to report absolute tensile strength or elongation with meaningful precision, so the measured columns record only a qualitative pass/fail check against the manufacturer's datasheet range, confirming that mixing ratios and cure conditions produced material behaviour consistent with specification. Formal ISO 37 characterisation on a calibrated universal testing machine would be required before any quantitative tensile values could be cited.

Table 3 – Preliminary Tensile Testing Results for Silicone Materials

Property	Ecoflex 00-30 (Measured)	Ecoflex 00-30 (Datasheet)	Dragon Skin 30 (Measured)	Dragon Skin 30 (Datasheet)
Shore Hardness	00-30*	00-30	A-30*	A-30
Tensile Strength	Pass (within datasheet range)	1.38 MPa	Pass (within datasheet range)	3.45 MPa
Elongation at Break	Pass (within datasheet range)	900%	Pass (within datasheet range)	364%
Tear Strength	–	38 pli	–	102 pli
Mixed Viscosity	–	3,000 cP	–	20,000 cP
Cure Time (23°C)	–	4 hours	–	16 hours

** Hardness confirmed qualitatively by comparison with calibrated Shore durometer samples. Quantitative tensile values are not reported because the available test rig is not calibrated to the precision that significant figures would imply; a pass/fail check against the manufacturer's datasheet range is reported instead (see Section 6.2 for discussion).*

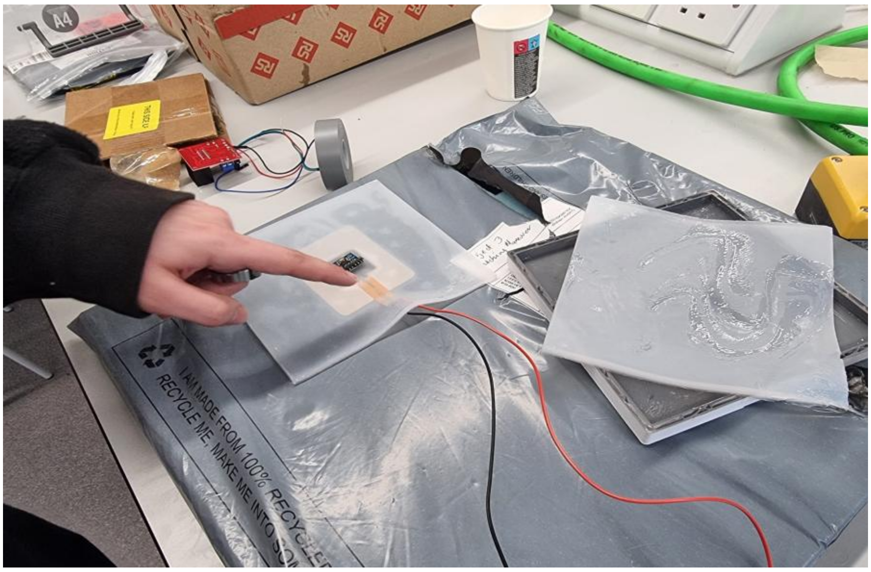

Figure 11 – Silicone test samples with embedded piezoelectric vibrator testing feedback through skin

5.3 Structural Integrity Testing

This section presents the deflection-test validation study used to inform the choice of skeletal filament. Four candidates (PLA, PETG-CF, ASA-CF and ABS) were compared on effective stiffness and elastic behaviour under representative loads, rather than on datasheet values, since printed parts must be characterised in the form they are used.

5.3.1 Experimental Method

Horizontal cantilever specimens of each material were clamped in a bench vice with a 3D-printed mount on the vice face setting a consistent clamp position. A 15 g carrier was hung from the free end via a fine cord, and masses were added incrementally (±1 g). Tip deflection was read against a ruler (±0.5 mm). All specimens were printed with identical parameters (0.2 mm layer height, 15% gyroid infill, 2 perimeter walls, nominal 5 × 4 mm rectangular cross-section) and free length ~251 mm.

Effective stiffness k was calculated by linear regression of applied force F against tip deflection δ, forced through the origin: $k = \frac{\Sigma(F \cdot \delta)}{\Sigma(\delta^2)}$. This uses the full loading curve rather than a single point and so is less sensitive to individual reading noise. Young's modulus was then recovered from Euler-Bernoulli beam theory:

$$E = \frac{k \cdot L^3}{3 \cdot I} \text{ where } I = \frac{b \cdot h^3}{12}$$

Uncertainty in E was propagated analytically through L, I and the regression standard error SE(k), with assumed measurement uncertainties δ ±0.5 mm, mass ±1 g, L ±1 mm, and b, h ±0.1 mm each. Only one specimen was tested per material; the implications are addressed in Section 6.2.

5.3.2 Regression Results

Table 4 – Regression-derived effective stiffness, apparent Young's modulus and uncertainty for each material.

Material	n	k (N/mm)	E (MPa)	u(E) (MPa)	δ/L max	E_app range (MPa)	Trend
PLA	9	0.0129	2,600	± 210	25%	2,547 – 2,971	Drift
ASA-CF	4	0.0128	2,360	± 190	30%	2,078 – 2,701	Scatter
PETG-CF	7	0.0126	2,310	± 180	25%	2,259 – 2,453	Flat
ABS	4	0.0106	1,950	± 160	36%	1,501 – 2,087	Drift

n is the number of loading steps in the regression; u(E) is the combined standard uncertainty. E_app range is the spread of per-step apparent E values, indicating elastic stability. Trend is the qualitative behaviour of apparent E as load increased.

Ranked by effective stiffness: PLA > ASA-CF ≈ PETG-CF > ABS. The first three overlap within their combined uncertainty bounds and cannot be distinguished on stiffness alone; ABS is distinctly less stiff.

5.3.3 Loading Curves and Stiffness Evolution

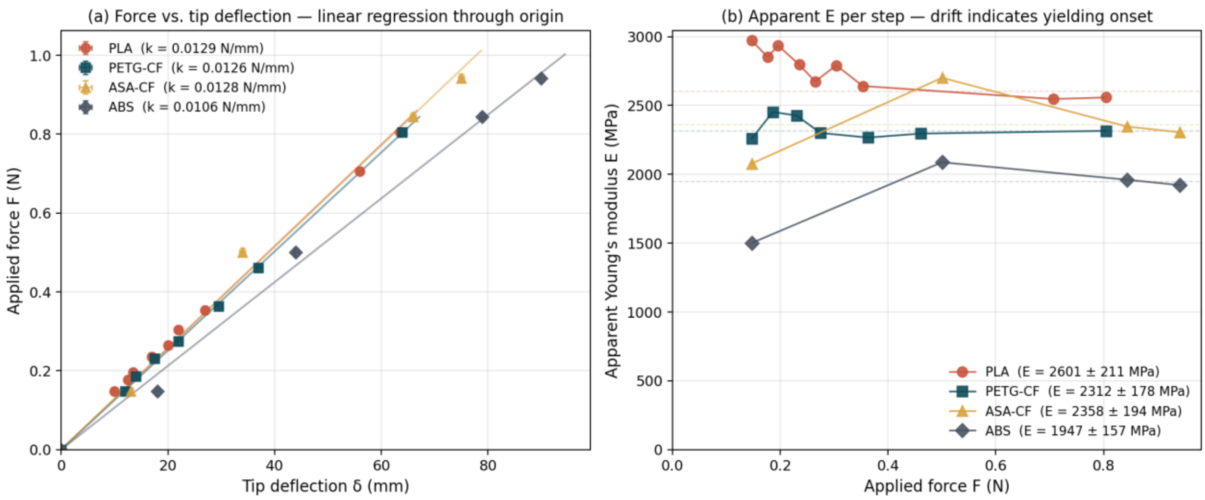

Figure 12 – (a) Applied force versus tip deflection for each material, with regression lines through the origin. Error bars show ±0.5 mm on δ and ±10 mg equivalent on F. (b) Apparent Young's modulus computed at each individual load step. Dashed horizontal lines mark each material's regression E. Flat curves (PETG-CF) indicate elastic stability; downward drift (PLA, ABS) indicates yielding onset.

Figure 10(b) is the more informative panel for material selection. PETG-CF's apparent E stays flat across the tested range (2,259 to 2,453 MPa, ±5% about its regression value), consistent with fully elastic behaviour. PLA drifts monotonically downward from 2,971 to 2,547 MPa, a ~15% reduction consistent with yielding onset; ABS shows similar drift at high loads. ASA-CF's first data point sits at the ruler resolution limit and should be treated with caution; the remaining points cluster around 2,300 to 2,700 MPa with no clear trend.

5.3.4 Supplementary Fixed-Load Observation (396 g Duct-Tape Roll)

A separate single-load test was performed using a duct-tape roll of measured mass 396 g, a load roughly four times larger than the maximum incremental mass reached during the regression tests. The specimen was loaded, observed, then unloaded. Observations are summarised in Table 5.

Table 5 – Qualitative observations under a single 396 g applied load.

Material	Observation
PLA	Whitening observed. Permanent deformation on unload.
PETG-CF	No whitening. Smallest permanent deformation of the four.
ASA-CF	No whitening. Mild permanent deformation on unload.
ABS	Strongest whitening observed. Permanent deformation on unload.

Whitening and permanent deformation are direct macroscopic indicators of plastic deformation, and the pattern mirrors the apparent-E drift: the materials whose E was stable (PETG-CF and, partially, ASA-CF) showed no whitening, and those whose E drifted (PLA and ABS) both whitened, with ABS, showing the steepest drift, whitening most visibly. The two measurements therefore support the same conclusion from independent evidence.

5.3.5 Material Selection for the Anatomical Core

Cast-application loading is cyclic and long-duration rather than a single extreme event, so the relevant material properties are elastic stiffness within the working range and freedom from accumulated plastic strain across many cycles; ultimate tensile strength is secondary. On stiffness alone the four candidates divide into a cluster (PLA, ASA-CF, PETG-CF at ~2,300 to 2,600 MPa) and ABS at ~1,950 MPa, with the cluster indistinguishable statistically given the single-specimen uncertainty. ABS, being both the least stiff and the earliest to yield, is not suitable for the structural skeleton and is retained only as a reference baseline.

Elastic stability is the deciding factor among the remaining three candidates. PETG-CF was the only material whose apparent Young's modulus did not drift under progressive load (Figure 10b), the only material showing no whitening under the 396 g load, and had the smallest permanent deformation. PLA drifted by ~15% across the regression range and whitened, indicating that realistic trainer loads would push it towards yield, with progressive sagging or softening developing over many cycles and damaging simulator fidelity well before outright fracture. ASA-CF is a reasonable second choice but its elastic stability cannot be confirmed from this test set alone because the low-load data sits at the ruler resolution limit.

The production specification is therefore PETG-CF for all load-bearing skeletal elements. The prototype distal radius was printed in ASA-CF as the closest-available substitute, with remaining bones in PLA as a filament-and-time compromise to be replaced in the production version (Section 7.2).

6. Discussion

6.1 Design Achievements

Modularity: met. The anatomical core successfully fulfils its role as a multi-functional integration chassis. The layered construction (skeleton, muscle, skin) allows individual layers to be replaced without rebuilding the whole model, supporting maintainability in a training environment where the skin degrades with repeated cast application. Internal routing accommodates all subsystem requirements declared by the team: tubing for pulsatile flow, recesses for piezoelectric actuators, sensor pockets, and wire channels to the central controller. No interface change was required in response to subsystem evolution during the project, which is the strongest evidence that the modularity target was met.

Fidelity: partially met. The dual-durometer silicone system is a significant advance over single-material competitors. In informal handling tests, users consistently identified the layered model as more realistic than uniform samples when palpating for bony landmarks through tissue, a finding consistent with multi-layer tissue simulation literature (Dixon et al., 2021; Favier et al., 2021). Fidelity has, however, been demonstrated only qualitatively: silicone tensile properties were confirmed at a pass/fail level (Table 3), no instrumented palpation comparison against a real or cadaveric forearm has been performed, and the fracture-site geometry is a simplified transverse break. The result is sufficient for a prototype but not yet qualified for clinical validation.

Durability: partially met. PETG-CF was selected on the cantilever study in Section 5.3 as the only candidate whose apparent modulus remained stable across the loading range. The prototype is built predominantly in PLA with an ASA-CF distal radius and is therefore expected to accumulate plastic strain sooner than a production unit; full durability validation has not been performed on either configuration and is scheduled with the upgraded cantilever programme in Section 7.2.

Cost: met. The estimated material cost is below the £500 target and well below the £2,000 price of the closest commercial comparator (Limbs & Things, 2019), directly supporting the goal of widening access.

6.2 Limitations and Critical Assessment

First, the physical prototype deviates from the production specification: PETG-CF for all load-bearing skeletal elements. Filament availability meant only the distal radius was printed in ASA-CF (the closest substitute on elastic stability), and the remaining bones in PLA. CAD geometry, wall count and infill pattern are unchanged between configurations; only filament and nozzle temperature differ, so the transition to PETG-CF requires only re-printing existing STLs. The prototype is functional for demonstration and initial integration, but the PLA components would accumulate plastic strain over repeated cycles and should not be deployed in live training without replacement.

Second, the tensile testing used a manual rig that lacks the precision of a calibrated Universal Testing Machine (UTM); the pass/fail results give reasonable confidence in material behaviour, but formal ISO 37 characterisation would be required for any clinical validation pathway.

Third, the cantilever study underpinning the PETG-CF selection has several limitations. Only one specimen was tested per material (N = 1), so u(E) in Table 4 reflects measurement precision only and gives no estimate of the print-to-print variability that dominates FDM scatter. The maximum tip-deflection ratio δ/L reached 25 to 36%, well beyond the ~10% validity limit of small-deflection beam theory, so absolute E values are systematically under-reported; the ranking between materials remains informative because all specimens were loaded into the same regime, but absolute values require a large-deflection correction. Additional sources of scatter include uncharacterised bench-vice compliance, inconsistent load-hold time (allowing variable creep), no failure test, and first-data-point uncertainty at the ±0.5 mm ruler resolution limit. Section 7.2 lists the follow-up tests needed to upgrade the study to a defensible qualification.

Fourth, the model has no mounting stand and must be held or supported on a surface during training, a practical limitation to be addressed with a clamp or suction-cup system.

6.3 Ethical Considerations

The simulator is deliberately designed without gender, race or body-specific characteristics: neutral grey silicone pigmentation and 50th-percentile anthropometric proportions (NASA, 1978) rather than any specific individual. This removes representation or bias concerns in medical training, consistent with the approach of established medical simulation companies.

Although this is not a medical device intended for patient contact, the design must still consider the duty of care owed to trainees: the system must not provide misleading feedback that instils incorrect technique. This is shared with the sensing and UI workstreams, but the anatomical core contributes by making palpation landmarks, tissue compliance and fracture mechanics physiologically plausible. Consultation with Dr Hasnain Abbasi (GP) confirmed the model’s tactile response was considered appropriate for training.

6.4 Environmental and Societal Impact

The environmental impact is moderate. PETG and PLA are recyclable thermoplastics, and failed prints can be reground and re-extruded. The platinum-cure silicones are not currently recyclable and will generate end-of-life waste, though the modular design means only degraded skin layers require replacement. Accessible, low-cost manufacturing (FDM printing, room-temperature silicone casting) means the simulator can be produced in any university workshop without specialised equipment.

Four mitigations are proposed. First, the modular skin-panel design roughly trebles the usable life of the artefact per unit of silicone by decoupling skin-layer lifetime from the skeleton. Second, platinum-cure silicone was specified over tin-cure: both are non-recyclable, but platinum systems contain no heavy-metal catalysts, so cured waste is non-hazardous under UK regulations (Environment Agency, 2021). Third, end-of-life PETG-CF components can be reground and redeployed as non-structural print stock. Fourth, an institutional skin-panel take-back scheme would consolidate silicone waste at the supplying workshop for controlled incineration via licensed streams, avoiding dispersed landfill disposal.

Societally, the project has the potential to improve patient outcomes by giving medical students more and better practical training before they encounter real patients. Reducing the cost barrier from ~£2,000 to ~£500 for the complete system brings simulation within reach of more institutions.

6.5 Risk Assessment

Table 6 – Risk Assessment for Anatomical Core Manufacture and Training Use

Hazard	Risk	Likelihood (1–10)	Mitigation
3D printer (hot nozzle/bed)	Thermal burns	3	Allow bed to cool below 40°C before handling. Use heat-resistant gloves.
Silicone mixing (Part A/B)	Skin irritation	4	Wear nitrile gloves and safety glasses. Work in ventilated area.
Sanding/deburring moulds	Cuts, dust inhalation	5	Use cut-resistant gloves. Wear dust mask. Use extraction.
Filler primer spray	Inhalation, flammability	5	Apply in well-ventilated area or spray booth. No open flames.
Sharp printed edges	Cuts	4	Deburr all parts before handling. File sharp edges.
Heat-set insert installation	Burns from soldering iron	5	Use appropriate soldering iron tips. Allow parts to cool.
Silicone skin tear during session	Exposure of internal components; possible pinch injury	6	Inspect skin between sessions; replace skin layer when tears exceed 10 mm using the modular glove design; cover any exposed rigid components during reassembly.
Fracture site seizes due to plaster or debris ingress	Trainee applies excessive force trying to free the joint, risking fracture of the skeletal part or injury to the trainee's hand	4	Clean the fracture site after each session with a soft brush; document a maximum hand-force limit in the instructor brief; design includes a relief gap so that debris can be flushed rather than accumulated.
Excessive force applied by trainee to the arm	Skeletal fracture of prototype parts; sharp edges exposed to trainee	5	Instructor brief includes a maximum-force limit calibrated to realistic cast-application forces; mounting base (to be added in future iterations) will include a compliant mechanical stop to limit applied torque before any skeletal part reaches its failure load.

7. Conclusions and Future Work

7.1 Conclusions

The following conclusions are drawn from this work: A full upper-limb anatomical model (humerus to fingertips) was designed and partially manufactured using accessible digital fabrication methods at a material cost under £500. The dual-durometer silicone system (Ecoflex 00-30 skin over Dragon Skin 30A muscle) produces a layered compliance profile that qualitatively replicates the non-linear tactile response of real forearm tissue during palpation. PETG-CF was identified as the optimal skeletal material through a cantilever deflection study of PLA, PETG-CF, ASA-CF and ABS, being the only material with a flat apparent Young's modulus across the loading range, no whitening under the 396 g load, and the smallest permanent deformation. It is therefore the production specification for all load-bearing skeletal elements. The prototype uses ASA-CF for the distal radius and PLA for the remaining bones as a filament-constrained fallback, to be replaced in production. The modular mould-based manufacturing approach allows individual tissue layers to be replaced independently, supporting long-term maintainability in a training environment. Internal routing for flow tubing, sensor wiring and haptic actuators was integrated into the skeletal and muscular geometry without compromising anatomical accuracy or structural integrity. Preliminary tensile testing, evaluated as a qualitative pass/fail against manufacturer datasheet ranges, confirmed that the cast silicone samples behaved consistently with specification.

7.2 Future Work

Several items of future work follow directly from the limitations identified in Section 6.2. The most immediate is re-printing the remaining skeletal components in PETG-CF to bring the prototype in line with the production specification. The cantilever validation study should be upgraded from an indicative screen to a defensible qualification: three specimens per material drawn from two print batches to capture batch variability; shorter specimens or thicker cross-sections to bring δ/L below the small-deflection limit; a 0.01 mm dial indicator in place of the ruler; a vice-compliance baseline against a rigid reference; a fixed load-hold time; a failure test; and a cyclic-loading test (~1,000 cycles at representative load) to validate permanent-set behaviour. Validation should then move to actual skeleton geometry, measuring distal-radius bending stiffness under a representative cast-squeeze load and confirming sensor-tolerance. Separately, ISO 37 tensile characterisation of the silicones on a calibrated UTM, a mounting stand, interchangeable fracture modules (including comminuted and Smith's variants), and long-term durability testing of the skin layer are all identified.

Acknowledgements

I want to extend my thanks to Dr Majid Taghavi for his supervision and guidance throughout the project. His advice and insights have been invaluable to the success of this project. My thanks are also extended to Dr Hasnain Abbasi for his clinical insights on fracture management and casting technique. The contributions of teammates Bohdan Skulimovskyi (sensing), Bartosz Bester (haptic feedback), Yuvraj Panesar (user interface), and Doanh Le (flow pumping) are gratefully acknowledged, particularly for their collaboration on interface specifications. The Queen Mary University of London workshop technicians provided essential support with access and materials.

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Appendix

A.1 Additional Figures

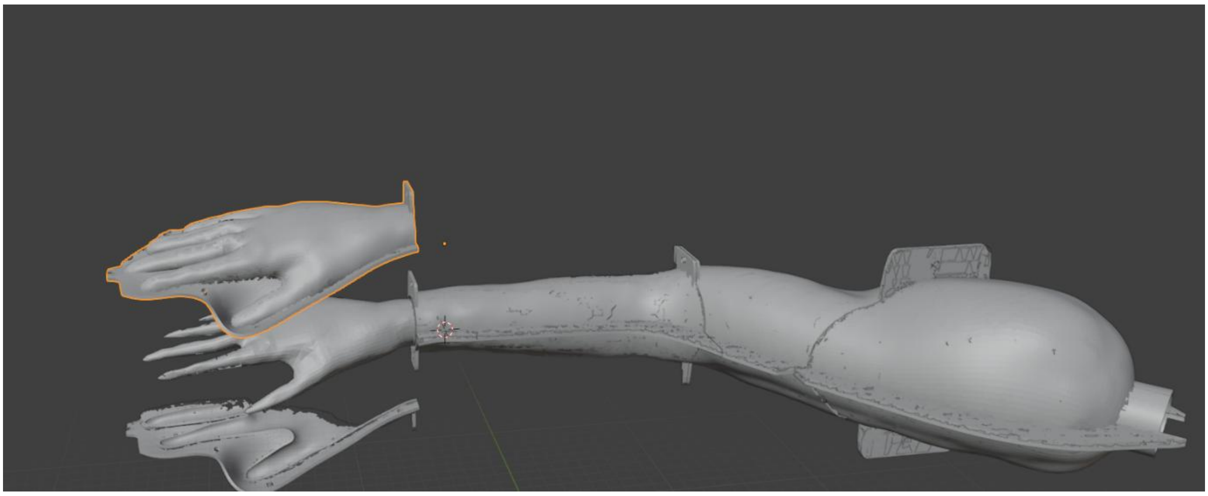

Figure A1 – Blender model of complete arm mould assembly showing hand mould and forearm sections